
When Moment Summaries Can Certify Attention

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Abstract

Moment summaries can avoid a token scan if their uncertainty cannot change the required output decision. We characterize this condition exactly for independent block intervals, derive exponential-attention bounds from projected signed moments, and tighten them using dependence between block mass and value range. A Lean development proves the real-arithmetic certificate chain; all 64 theorems build and pass an axiom audit. Exact-rational experiments certify 61 of 149 coordinates with value coupling versus 57 with boxes, including one controlled gain beyond a global value hull. On a GH200, parallel block reduction reduces a favorable screening stage from 118.65 to 20.09 μ s, but the complete path remains slower than fused dense attention; trained-model traces yield no passes in 2,016 rank-eight head instances. These results identify two obstacles to useful certification: exponential bound inflation and decision/fallback overhead.

1. Introduction

An approximate attention value and a certified output decision solve different problems. A small absolute error can still cross a rounding boundary, while a coarse value estimate may determine a broad downstream decision exactly. The relevant quantity is therefore the uncertainty relative to the consumer’s decision boundary.

We study positive normalized reductions represented by block masses and centered value moments. At any boundary c , the normalized comparison reduces to the sign of an unnormalized residual $N - cZ$. Independent interval metadata gives attained extrema of this residual. If total mass is positive throughout the box, two strict sign tests are necessary and sufficient for every compatible output to lie in a given open cell. A rejection identifies information missing from the box; it need not identify an inaccurate candidate value.

[Code, proofs, and measurements.](#)

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We test which dependence information enables certification, which error term prevents it on trained activations, and whether avoided token work repays construction, screening, and fallback. The evidence combines value-coupled bounds, a formal real-attention proof chain, and GH200 implementation ablations.

Value coupling adds certificates, although a global value hull explains most gains. Trained activations expose discarded-score uncertainty and full-rank Taylor inflation. Parallel reduction removes a GPU bottleneck, yet the complete path remains slower than fused dense attention throughout the sweep.

Scope. Lean proves the real-arithmetic chain under concrete centering, radius, symmetry, and row-sum witnesses. The rational checker and the directed CUDA checker are tested implementations, not extracted proofs. Ordinary NumPy/CUDA summaries do not provide outward intervals. Exact-real rounding also differs from the accumulation contract of a particular fused GPU kernel. Appendix B maps these claims to the checked theorems and remaining implementation bridges. Code, measurements, and reproduction commands are available in the [research repository](#).

Related work. Adaptive predicates refine uncertain signs (Shewchuk, 1997), while Taylor models retain polynomial structure with bounded remainders (Berz & Hoffstätter, 1998). Multipole attention combines grouped interactions and selected exact work (Hooper et al., 2025; Kang et al., 2023); TaylorShift and symmetry-aware features reorganize polynomial attention (Nauen et al., 2024; Heinsen & Kozachkov, 2026). COBS uses covariance statistics for block selection, SPLA combines sparse and residual linear paths, and LOCKS studies limits of moment summaries for broad scores (Tian et al., 2026; Wang et al., 2026; Hwang, 2026). WitCert and attention-memory observability contracts already connect runtime witnesses, formal scope, and fallback (Wei et al., 2026a;b). Our focus is the consumer-specific residual interface, the projected signed moments supplying it, and stronger mass/value dependence. These general principles are established; comparisons with unreproduced systems concern method scope, not measured performance.

2. Certificates at Observation Boundaries

Partition a finite, nonempty visible input into nonempty blocks $b = 1, \dots, B$. A positive normalized reduction has coordinates

$$y_j = N_j/Z, \quad N_j = \sum_b n_{bj}, \quad Z = \sum_b z_b > 0. \quad (1)$$

For fixed centers ν_{bj} , write $m_{bj} = n_{bj} - \nu_{bj}z_b$ and assume sound metadata

$$0 \leq l_b \leq z_b \leq u_b, \quad m_{bj}^- \leq m_{bj} \leq m_{bj}^+. \quad (2)$$

All quantities must describe the same input, mask, query, scaling, and data version. Positivity of Z is a separate premise; $\sum_b l_b > 0$ suffices.

An *observation cell* (a_j, b_j) is an open interval on which the consumer is constant. Proving y_j belongs to this cell therefore determines its observation. For finite numerical BF16 rounding, neighboring representable values supply midpoint boundaries; strict containment avoids ties. This contract does not distinguish positive and negative zero by their bits. Alternatively, a monotone consumer Q satisfies $Q(l) = Q(u) \Rightarrow Q(y) = Q(l)$ whenever $l \leq y \leq u$. A concrete IEEE-754 model and its monotonicity theorem are not supplied.

2.1. Exact Certificates for Independent Boxes

At a boundary c , define

$$\begin{aligned} \mathcal{L}_j(c) &= \sum_b m_{bj}^- \\ &\quad + \sum_b \min\{(\nu_{bj} - c)l_b, (\nu_{bj} - c)u_b\}, \end{aligned} \quad (3)$$

$$\begin{aligned} \mathcal{U}_j(c) &= \sum_b m_{bj}^+ \\ &\quad + \sum_b \max\{(\nu_{bj} - c)l_b, (\nu_{bj} - c)u_b\}. \end{aligned} \quad (4)$$

Theorem 2.1 (Boundary certificate). *Under Equations (1) and (2),*

$$\mathcal{L}_j(c) \leq N_j - cZ \leq \mathcal{U}_j(c). \quad (5)$$

Consequently,

$$\begin{aligned} \mathcal{L}_j(a_j) &> 0, \quad \mathcal{U}_j(b_j) < 0 \\ \implies a_j &< y_j < b_j. \end{aligned} \quad (6)$$

Proof. The centered residual identity is

$$N_j - cZ = \sum_b [m_{bj} + (\nu_{bj} - c)z_b]. \quad (7)$$

For any scalar α , the extrema of αz on $[l, u]$ are $\min(\alpha l, \alpha u)$ and $\max(\alpha l, \alpha u)$. Apply these bounds to each summand and sum. Dividing the strict residual signs by $Z > 0$ gives the cell inequalities. $\square$

The certificate requires no division. A numerical quotient may choose a candidate cell: a poor choice reduces acceptance but cannot invalidate sound residual signs.

Theorem 2.2 (Optimality for independent metadata). *Over the Cartesian product of the intervals in Equation (2), $\mathcal{L}_j(c)$ and $\mathcal{U}_j(c)$ are the attained minimum and maximum of the residual at each fixed c .*

The proof is given in Appendix A.1.

This optimality concerns the stated product box. Actual masses and numerators can be correlated, and the two extrema need not come from the same input. Residual optimality needs no positive lower bound on total mass; interpreting every extremizer as a normalized output additionally requires $\sum_b l_b > 0$.

Corollary 2.3 (Sufficiency of box metadata). *If $\sum_b l_b > 0$, every compatible box output belongs to (a_j, b_j) if and only if $\mathcal{L}_j(a_j) > 0$ and $\mathcal{U}_j(b_j) < 0$.*

Proof. Sufficiency follows from Theorem 2.1. If the lower test fails, Theorem 2.2 supplies a vertex with residual at a_j at most zero. Its denominator is positive, so its output is at most a_j . The upper extremizer gives the corresponding counterexample at b_j . $\square$

A rejection thus means that the metadata admits a conflicting output, even if the original tokens cannot realize that vertex. Acceptance then needs tighter bounds, retained dependence, or a different observation contract.

The two-block example in Example A.1 and Figure 5 shows a strict improvement over a symmetric midpoint residual bound.

2.2. Refinement and Retained Value Dependence

Theorem 2.4 (Monotone refinement). *Replacing each interval by a nonempty subinterval cannot decrease $\mathcal{L}_j(c)$ or increase $\mathcal{U}_j(c)$ at a fixed boundary c . Intersecting two sound enclosures preserves the truth.*

Proof. A minimum over a smaller feasible set cannot decrease, and a maximum cannot increase. Apply this fact to every block and sum. A point contained in both enclosures belongs to their intersection. $\square$

Selected blocks may therefore be refined without discarding earlier work. The invariant is sound enclosure; the implementation's uncertainty priority is a heuristic. Monotonicity applies to a fixed cell. Refinement need not terminate with a strict certificate at a rounding midpoint, so execution requires bounded work and a tie-aware path or declared reference fallback.

For nonnegative block weights, exact centered value extrema $p_{bj} = \min_{i \in b} (v_{ij} - \nu_{bj})$ and $q_{bj} = \max_{i \in b} (v_{ij} - \nu_{bj})$ imply

$$p_{bj} z_b \leq m_{bj} \leq q_{bj} z_b : \quad (8)$$

multiply each value inequality by its weight and sum. Retain this dependence by defining the nonempty compact polygon

$$P_{bj} = \left\{ \begin{array}{l} l_b \leq z \leq u_b, \\ (z, m) : m_{bj}^- \leq m \leq m_{bj}^+, \\ p_{bj} z \leq m \leq q_{bj} z \end{array} \right\}. \quad (9)$$

Theorem 2.5 (Coupled certificate). *In Equations (3) and (4), replace each block term by the minimum or maximum, respectively, of $m + (\nu_{bj} - c)z$ over P_{bj} . The resulting residual enclosure and cell certificate are sound, never wider than the box bounds, and improve monotonically under sound box intersection with fixed value extrema.*

Proof. The true pair belongs to P_{bj} by Equations (2) and (8). The residual identity remains unchanged. Each polygon is contained in its rectangle, and shrinking the rectangle shrinks its intersection with the fixed cone. The extrema can only improve; summation and $Z > 0$ finish the proof. $\square$

The rational optimizer evaluates these extrema at polygon vertices using a constant number of rational operations per block/channel; bit cost depends on the fractions. The complete enumeration argument is in Appendix A.5. Inconsistent metadata is rejected.

Joint optimization can be strictly tighter than intersecting separate scalar bounds; Example A.2 gives an exact example.

Lean proves the positive-weight coupling and box certificate. The polygon optimizer has the argument in Appendix A.5 and independent exact geometry tests; its executable correctness is not formalized.

A sufficient margin-versus-width condition explains difficult near-midpoint coordinates and eventual certification under convergent sound refinement (Proposition A.3).

3. Projected Signed-Moment Enclosures

Attention has the exact-real form

$$y_j(q) = \frac{\sum_i e^{q^T k_i} v_{ij}}{\sum_i e^{q^T k_i}}. \quad (10)$$

The query includes attention scaling; keys are those attended to after positional transformations. Let μ_b, ν_b be exact block means and set $\delta_i = k_i - \mu_b$, $x_i = v_i - \nu_b$. Select $r \leq d$ coordinates and write

$$\begin{aligned} q^T \delta_i &= t_i + e_i, & t_i &= u^T \delta_i^{(r)}, \\ |t_i| &\leq \rho_b, & |e_i| &\leq \epsilon_b. \end{aligned} \quad (11)$$

The implementation selects a coordinate prefix; permutation permits another fixed subset. No approximate PCA basis is substituted. Stored coordinate radii $a_{bk} \geq \max_{i \in b} |\delta_{ik}|$ and value radii $R_{bj} \geq \max_{i \in b} |x_{ij}|$ give

$$\rho_b = \sum_{k < r} |q_k| a_{bk}, \quad \epsilon_b = \sum_{k \geq r} |q_k| a_{bk}. \quad (12)$$

3.1. Signed Moments and a Quadratic Witness

Using block means, store

$$\begin{aligned} \Sigma_b &= \text{mean}(\delta^{(r)} \delta^{(r)T}), \\ C_b &= \text{mean}(\delta^{(r)} x^T), \\ H_{bj} &= \text{mean}(\delta^{(r)} \delta^{(r)T} x_j). \end{aligned} \quad (13)$$

The full signed tensor H is temporary construction data. Query metadata retains its diagonal D_{bj} and a witness

$$\eta_{bj} \geq \max_a \sum_k |(H_{bj} - D_{bj})_{ak}|. \quad (14)$$

Here D denotes either the diagonal matrix or its stored diagonal. Choosing $D = 0$ is also valid. For $r = 0$, the empty row-sum witness is zero.

Lemma 3.1 (Quadratic witness). *If F is symmetric and $\eta \geq \max_a \sum_k |F_{ak}|$, then $|u^T F u| \leq \eta \|u\|_2^2$.*

The proof is given in Appendix A.2.

Retaining the diagonal weakly reduces the row-sum witness but shifts the center; intervals need not be nested or improve acceptance (Appendix A.9).

3.2. A Sharp Exponential Remainder

Define the nonnegative coefficient

$$\kappa(\rho) = \begin{cases} (e^\rho - 1 - \rho - \rho^2/2)/\rho^2, & \rho > 0, \\ 0, & \rho = 0. \end{cases} \quad (15)$$

Lemma 3.2 (Sharp quadratic majorant). *For $\rho \geq 0$ and $|t| \leq \rho$,*

$$\begin{aligned} |e^t - 1 - t - t^2/2| &\leq \kappa(\rho) t^2, \\ \kappa(\rho) &\leq e^\rho \rho/6. \end{aligned} \quad (16)$$

For $\rho > 0$, the second inequality is strict and $\kappa(\rho)$ is the smallest uniform quadratic coefficient on $[-\rho, \rho]$.

The proof is given in Appendix A.3.

At radius zero, zero is the chosen nonnegative coefficient; minimality over arbitrary real coefficients is not claimed. Near zero, numerical evaluation uses a series to avoid cancellation. It is not outward without a bounded remainder; the rational path instead uses an upper enclosure of e^ρ . Lean

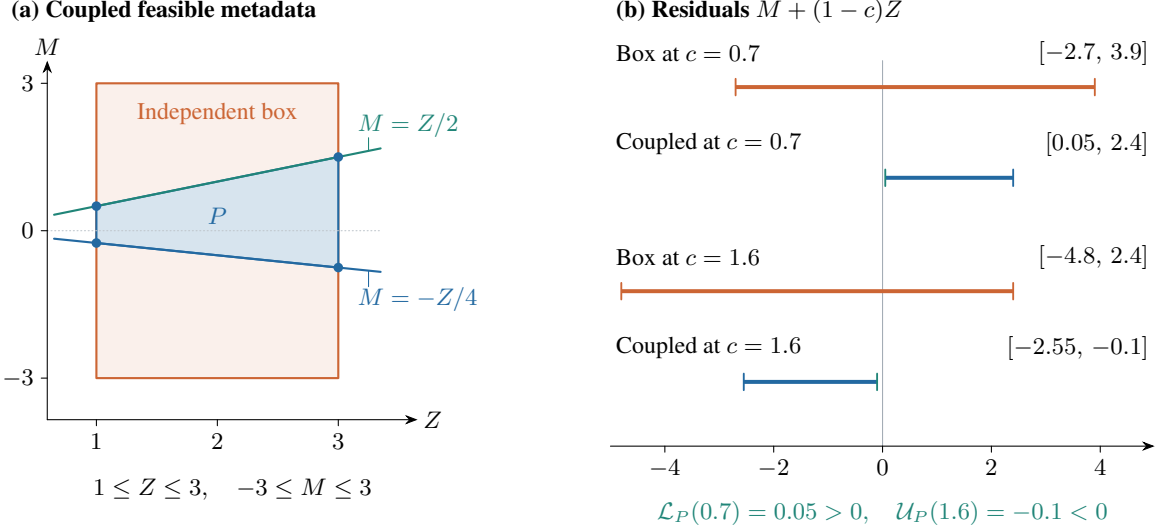


Figure 1. An analytic coupled-geometry example: value dependence removes box corners that obstruct both boundary signs. The target interval is illustrative.

derives the bound from the convergent real exponential series and discharges the remainder premise of the subsequent lifting lemma.

The remainder and discarded-score inflation retain exponential growth (Figure 6).

3.3. From Moments to Full-Score Blocks

Choose any common finite shift s and define the actual quantities

$$\begin{aligned} z_b(s) &= \sum_{i \in b} e^{q^T k_i - s}, \\ m_{bj}(s) &= \sum_{i \in b} e^{q^T k_i - s} x_{ij}. \end{aligned} \quad (17)$$

The factor e^{-s} cancels from Equation (10); all following bounds and residuals concern these shifted quantities. Set

$$\begin{aligned} a_b &= q^T \mu_b, & w_b &= n_b e^{a_b - s}, \\ \sigma_b^2 &= u^T \Sigma_b u, & A_b &= 1 + \sigma_b^2 / 2, \\ \tau_b &= \kappa(\rho_b) \sigma_b^2. \end{aligned} \quad (18)$$

$$\begin{aligned} \tilde{l}_b &= w_b \max\{1, A_b - \tau_b\}, \\ \tilde{u}_b &= w_b (A_b + \tau_b), \end{aligned} \quad (19)$$

$$\begin{aligned} \hat{m}_{bj} &= w_b \left[(C_b^T u)_j + \frac{1}{2} u^T D_{bj} u \right], \\ \beta_{bj} &= w_b \left[\frac{1}{2} \eta_{bj} \|u\|_2^2 + \tau_b R_{bj} \right]. \end{aligned} \quad (20)$$

Here n_b is the block cardinality.

Theorem 3.3 (Full-score block enclosures). *Under the centering, radius, symmetry, and row-sum conditions above, sound bounds for Equation (17) are*

$$l_b = e^{-\epsilon_b} \tilde{l}_b, \quad u_b = e^{\epsilon_b} \tilde{u}_b, \quad (21)$$

$$m_{bj}^\pm = \hat{m}_{bj} \pm [\beta_{bj} + (e^{\epsilon_b} - 1) \tilde{u}_b R_{bj}]. \quad (22)$$

The proof is given in Appendix A.4.

Combining Theorems 2.1 and 3.3 certifies the original finite per-token quotient. The Lean witness supplies centering, radii, symmetry, and row sums; it does not assume the Taylor remainder or output enclosure. Lean uses unnormalized finite-sum tensors: multiply this paper’s H, D, η by n_b to obtain that representation. Matching the score decomposition and partition to the caller’s data remains necessary; the theorem is not a Python summary validator.

Both endpoint ranks are included. At $r = 0$, retained centered moments vanish and all score variation is discarded; at $r = d$, $\epsilon_b = 0$ eliminates projection inflation. No fixed rank is assumed useful on arbitrary trained activations.

3.4. Rational Evaluation and Outward Export

The rational implementation uses exact fractions and bounded exponential series. For $0 \leq x \leq 1/2$, a partial sum through degree n has first omitted term $a_{n+1} = x^{n+1}/(n+1)!$. Subsequent term ratios are bounded by $q = x/(n+2) < 1$, so the tail is at most $a_{n+1}/(1-q)$. The partial sum and that sum plus the tail enclose e^x . Positive interval squaring handles range reduction; reciprocal intervals handle negative arguments. Unsupported inputs are rejected. A separate rational exponential-sum oracle uses no

moment formulas; refinement intersects its result with the existing enclosure. These expensive routines are correctness artifacts, not the proposed fast GPU implementation.

Export converts rational endpoints outward to binary64 and compares each float’s exact rational value with its source. Centers also receive intervals because rational means may be unrepresentable. The host checker expands basic operations by one ULP; the CUDA draft uses directed rounding. Their validity still requires the stated floating-point semantics and faithful compilation. Both have been compiled and run on GH200, including memory checking of the imported CUDA path; neither is extracted from Lean. Ordinary NumPy/CUDA moment estimates do not become sound metadata through directed rounding of the final residual sum. Sound inputs remain a premise.

4. Representation and Execution Cost

For B blocks, key dimension d , value dimension h , and retained rank r , the base representation stores

$$B(2d + r^2 + 2rh + 3h + 1) \quad (23)$$

scalars. Coupling adds $2Bh$ value extrema. These counts exclude original K/V, index maps, source identities, scratch, and allocations. Original K/V remains available for fallback.

A block-scalar pass evaluates centroid scores, variance, radii, and mass terms once per query/block. A channel pass contracts the value moments, then a reduction tests the two boundaries. The arithmetic cost is $O(B(d + r^2 + rh + h))$, versus $\Theta(N(d + h))$ for a direct scan. The current CPU builder scans K/V and constructs a temporary signed tensor in $O(N(d + h + r^2h))$ work. These are word-operation counts; they omit rational bit complexity and do not imply a speedup. Clustering would add a setup stage.

Proposition 4.1 (Reuse threshold). *Let setup cost be $S \geq 0$, dense-query cost $D > 0$, full-output pass probability p , and expected screen/decision/return cost $C \geq 0$. If rejection executes the full dense path, expected cost for $R > 0$ reuses is*

$$T(R) = S + R[C + (1 - p)D]. \quad (24)$$

It is below RD exactly when both conditions hold:

$$pD > C, \quad R > \frac{S}{pD - C}.$$

Proof. Subtract RD and rearrange $S < R(pD - C)$. $\square$

This fixed-cost model makes the role of pass probability explicit. A heterogeneous scheduler needs its conditional costs. Our complete measured path is already slower than dense per query, so no finite reuse count amortizes its setup.

The executor validates the supported domain and source identity, selects a candidate cell, tests its boundaries, and refines or falls back under a work cap. Exact-rational refinement intersects sound intervals; numerical CUDA correction supplies estimates. Python fingerprints bind refinement to ordered visible K/V, query, and shift. They do not implement concurrent cache epochs or a serving memory manager. The measured numerical kernels compare serial, shared-scalar, and parallel block reductions; only the last changes reduction order. The directed imported-box checker has a separate contract.

5. Experimental Evidence

We test coupling, trained-activation uncertainty, and GPU execution separately. Appendix C.1 records arithmetic and implementation checks, including an executable interval defect found and repaired.

5.1. Coupling and Localized Uncertainty

Coupling accepts four more initial coordinates than the box certificate, but the global value hull explains three of those gains (Table 2 and Figure 2). The exact-rational study uses seed 962605, four retained ranks, four key spreads, narrow and broad value channels, exact midpoint channels, and named geometry and attention witnesses. All 53 cases, 149 initial coordinates, and sequential refinement stages are retained in [coupling artifact](#), including construction and actual rational fallback costs.

If all values lie in one rounding cell, positivity of attention weights already certifies the output. The remaining prescribed witness instead places keys $-2, 2$ with values $1 - 1/2560, 1 + 1/2560$ in one rank-zero block and a singleton key -16 with value 3, at $q = 1$. The outlier defeats the global hull, but its small block mass lets the coupled test certify BF16 value 1. An independent direct rational exponential sum verifies the certificate. This witness isolates the interaction between local value range and mass; it does not estimate coverage on trained models.

A numerical CPU experiment with $N = 8192$, $d = 16$, $h = 8$, $B = 16$, $Q = 24$, and one BLAS thread demonstrates when selective correction helps. The generator provides favorable four-coordinate structure. Tight blocks pass all 24 queries without correction; introducing one broad block makes every initial query fail, but correcting that block evaluates attention weights and values for only 1/16 of the tokens. Broadening every block requires correcting every token (Figure 7). The correction-token counter excludes the Python executor’s full-source input validation and fingerprint passes, which read K/V even without correction. CPU query times include these reads and executed correction or fallback; construction is recorded separately.

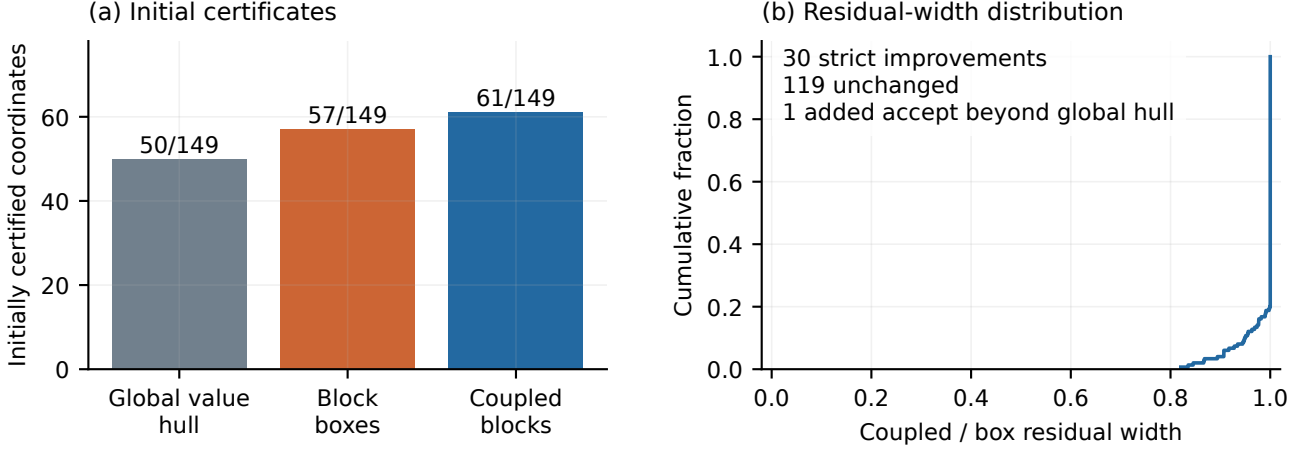


Figure 2. Coupling acceptance and residual widths for all initial coordinates, retaining unchanged widths and refusals. The global-hull baseline explains most additional accepts.

Every complete Python path is slower than dense NumPy: the experiment measures uncertainty localization, not GPU speed.

The GPU controls show why coordinate coverage alone is insufficient. Moderate inputs pass 10,197 of 19,008 coordinates across repeated rank configurations, but none of 297 complete outputs. A centered-value control passes 1,986 of 2,048 coordinates and only 6 of 32 outputs. Every coordinate must pass before an output is accepted; any failure triggers the measured full-batch fallback. Appendix C.3 retains the control protocol and strict midpoint refusals.

5.2. Trained Post-RoPE Activations

Neither trained activation set produces a numerical coordinate or whole-output pass. We capture post-RoPE Q/K/V from Qwen2.5-0.5B (Qwen Team, 2024) on prose, code, and arithmetic probes at prefix lengths 1,024 and 4,096. Rank eight covers 24 layers and 14 query heads: 2,016 head instances and 129,024 coordinates. Full rank in layers 0, 12, and 23 adds 252 head instances. Appendix C.4 specifies the model revision and capture protocol.

At rank eight, $e^\epsilon - 1$ amplifies centered-value uncertainty far beyond cell margins. Full rank removes that term, but the broad retained radius makes $\kappa(\rho)$ large (Table 3 and Figure 9). Median maximum absolute candidate error is approximately 0.82 at rank eight and 0.50 at full rank. Increasing rank therefore exchanges projection inflation for Taylor inflation while leaving errors much larger than the narrowest cells. Figure 3 expresses the competing terms in output units. Retained coordinates are a prefix of the actual transformed basis; no learned basis, PCA fitting, or clustering was used. Other representations remain untested, but must address within-block score variation, radius-bound looseness, and value dependence together.

5.3. GH200 Performance and Complete Costs

We measure 57 synthetic and nine trained prose trace/rank configurations on one NVIDIA GH200. The forced PyTorch fused FlashAttention SDPA baseline (Dao et al., 2022) matches BF16 inputs, scale, visible keys, and zero dropout. Seven samples of 20 invocations provide CUDA-event and host-wall times. Complete eager timings include screening, device decision reduction, CPU synchronization, and output cast or actual full-batch fallback; resident measurements exclude separately recorded setup. Appendix C.5 gives the environment, batch semantics, and sampling details; Figure 10 reports resident and complete-path latency.

Parallel block reduction improves the long-context, single-query resident kernel, while small cases retain adverse results (Table 4 and Figure 4). Originally each output coordinate occupies one thread looping serially over blocks; one query thus exposes only 64 threads. Sharing repeated scalar arithmetic barely helps. Assigning a CTA per coordinate exposes block parallelism, reducing the isolated reduction from $99.38 \mu\text{s}$ to $6.65 \mu\text{s}$ at $N = 32768, Q = 1$. With more queries or fewer blocks, additional CTAs and synchronization provide less benefit and can hurt. Isolated phase measurements diagnose this bottleneck; their sum does not replace measured complete latency (Figure 11).

Even the favorable $N = 32768, Q = 1, r = 4$ case loses end to end. Its resident parallel screen takes $20.09 \mu\text{s}$ under fixed-input graph replay versus dense attention’s $31.30 \mu\text{s}$; the complete eager path takes $83.11 \mu\text{s}$ versus $35.62 \mu\text{s}$, plus about 51.50 ms of reusable summary/workspace setup. Every measured complete path is slower than dense, so no finite reuse count repays setup under these costs. Appendix C.6 retains the footprint exclusions, phase measurements, and amortization results.

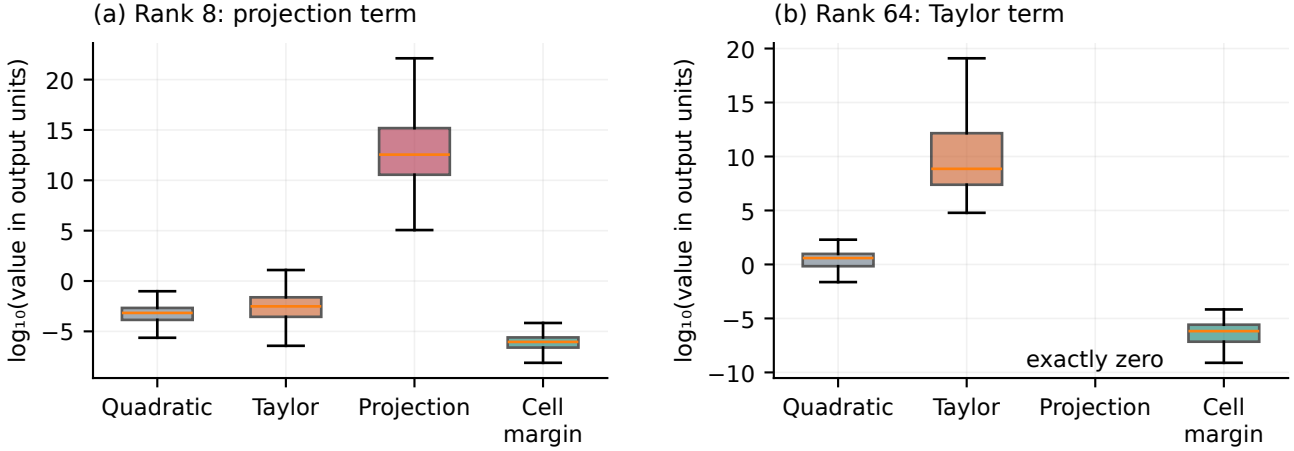


Figure 3. Maximum-channel omitted-quadratic, Taylor, and projection terms, divided by true shifted mass, compared with minimum-channel cell margins. These explain scales rather than additively decompose output error: mass uncertainty also contributes. JSON artifacts retain outliers beyond whiskers.

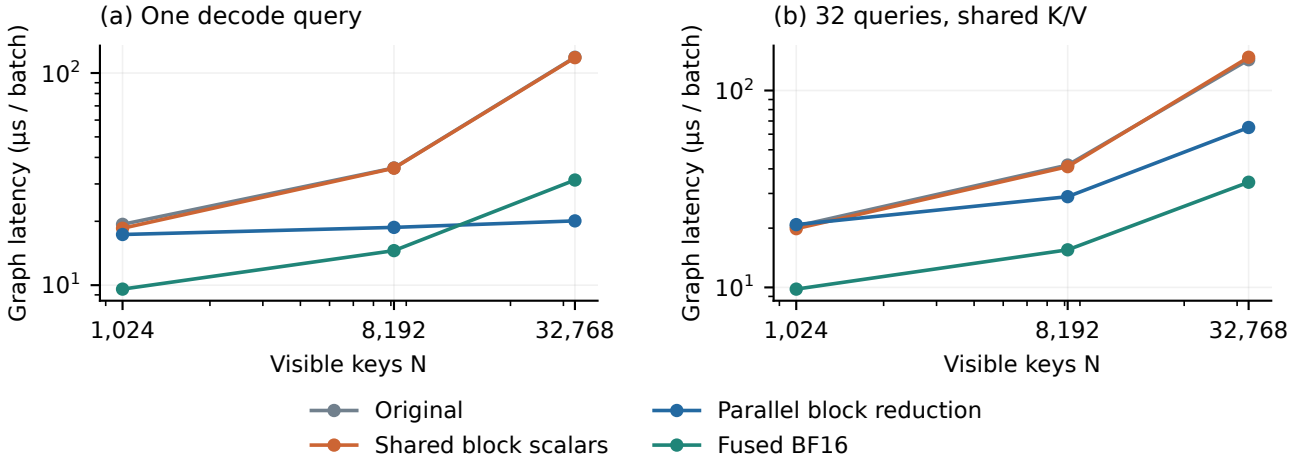


Figure 4. Three CUDA implementations under matched graph replay. Sharing scalar work leaves serial block reduction dominant; parallelizing blocks helps long-context single-query inputs, with synchronization costs at small N .

Across all 66 configurations, accepted numerical coordinates have zero observed BF16 mismatches against binary64 attention but 1,691 against the fused BF16 baseline. Repeated configurations make these aggregate counts dependent. Real-attention rounding and fused intermediate arithmetic are different targets; the binary64 comparison is itself numerical, and the summary path is not outward interval metadata for the directed checker.

A second run at committed implementation `d2c3668` reproduces all 66 cases with unchanged coverage and mismatch counts. Parallel-screen graph medians change by -1.08% to $+1.42\%$, with median ratio 1.002; every complete path remains slower than dense. Figures and tables retain the first sweep. Appendix C.7 provides provenance, host-wall variation, and the retained failed parity checks.

6. Discussion

Better certificates need information that controls the decision boundary. Test block partitions, tighter radii, higher-order residuals, and consumer margins separately, charging any query-dependent scan. Coupled certification on trained traces remains unmeasured.

Faster execution needs cheaper decisions and bounded fall-back work, together with a sound GPU summary builder. The reduction ablation identifies a useful device optimization; complete-path costs constrain its value. Our PyTorch baseline does not exhaust Hopper and decode implementations (Shah et al., 2024; Sanovar et al., 2024). The contribution is a proved real-arithmetic certificate and an empirical account of where its information and execution costs prevent acceleration.

A. Supporting Certificate and Moment Proofs

A.1. Box Optimality

Proof of Theorem 2.2. For the minimum, choose $m_{bj} = m_{bj}^-$ and $z_b = l_b$ if $\nu_{bj} - c \geq 0$, or $z_b = u_b$ otherwise. These simultaneous admissible choices minimize every summand in Equation (7). Opposite mass endpoints and $m_{bj} = m_{bj}^+$ attain the maximum. The matching inequalities are Equation (5). $\square$

A.2. The Row-Sum Quadratic Bound

Proof of Lemma 3.1. The inequality $|u_a u_k| \leq (u_a^2 + u_k^2)/2$ and symmetry give

$$\begin{aligned} |u^T F u| &\leq \frac{1}{2} \sum_{a,k} |F_{ak}| (u_a^2 + u_k^2) \\ &= \sum_a u_a^2 \sum_k |F_{ak}| \leq \eta \sum_a u_a^2. \end{aligned}$$

No numerical eigendecomposition is needed. $\square$

A.3. The Sharp Exponential Majorant

Proof of Lemma 3.2. If $\rho = 0$, then $t = 0$. Otherwise absolute convergence of the exponential series yields

$$\left| \sum_{n \geq 3} \frac{t^n}{n!} \right| \leq t^2 \sum_{n \geq 3} \frac{\rho^{n-2}}{n!} = \kappa(\rho) t^2.$$

Equality at $t = \rho > 0$ forces every uniform coefficient to be at least $\kappa(\rho)$. Substituting $n = m + 3$ and using $(m + 3)! \geq 6m!$ gives

$$\kappa(\rho) = \sum_{m \geq 0} \frac{\rho^{m+1}}{(m+3)!} \leq \frac{\rho}{6} \sum_{m \geq 0} \frac{\rho^m}{m!}.$$

For $\rho > 0$, terms with $m \geq 1$ make the inequality strict. $\square$

A.4. Full-Score Enclosure from Signed Moments

Proof of Theorem 3.3. Centering gives $\text{mean}(t_i) = 0$, $\text{mean}(t_i^2) = \sigma_b^2$, and $\text{mean}(x_{ij}) = 0$. Writing $r_i = e^{t_i} - 1 - t_i - t_i^2/2$, Lemma 3.2 gives $\text{mean}|r_i| \leq \tau_b$. Thus the projected mass lies in $w_b[A_b - \tau_b, A_b + \tau_b]$. Jensen's inequality also bounds it below by w_b , proving Equation (19).

The projected centered numerator equals

$$w_b \left[(C_b^T u)_j + \frac{1}{2} u^T H_{bj} u + \text{mean}(r_i x_{ij}) \right].$$

Apply Lemma 3.1 to $H_{bj} - D_{bj}$ and use $|\text{mean}(r_i x_{ij})| \leq \tau_b R_{bj}$ to obtain Equation (20).

Restoring discarded coordinates multiplies each positive projected weight by $e^{\epsilon_i} \in [e^{-\epsilon_b}, e^{\epsilon_b}]$, proving Equation (21). Since $|e^{\epsilon_i} - 1| \leq e^{\epsilon_b} - 1$, the centered numerator changes by at most

$$\begin{aligned} (e^{\epsilon_b} - 1) R_{bj} \sum_{i \in b} e^{a_b - s + t_i} \\ \leq (e^{\epsilon_b} - 1) \tilde{u}_b R_{bj}. \end{aligned}$$

Adding this perturbation proves Equation (22). $\square$

A.5. Exact Polygon Support

The rational optimizer enumerates polygon vertices. Candidate masses are l_b, u_b and $m_{bj}^-/s, m_{bj}^+/s$ for each nonzero $s \in \{p_{bj}, q_{bj}\}$. At a feasible mass z , the centered endpoints are $\max(m_{bj}^-, p_{bj}z)$ and $\min(m_{bj}^+, q_{bj}z)$. These cover

intersections of vertical, horizontal, and sloping boundaries. Distinct sloping lines meet at $z = 0$, necessarily a mass endpoint if feasible under $l_b \geq 0$; coincident lines and degenerate segments add no candidates. A linear objective attains its extrema at these vertices. Empty candidate sets indicate inconsistent metadata and are rejected. The number of rational operations is constant per block/channel; their bit cost is not.

A.6. Boundary Geometry

Example A.1. Two blocks with centers 0, 2, masses in $[1, 2]$, and zero centered numerators have output range $[2/3, 4/3]$. Boundary signs certify $(0.6, 1.4)$. A symmetric residual bound about 1 using midpoint masses has radius 0.5 and fails. This is a strict improvement in this example, not a dominance claim over every correlated certificate.

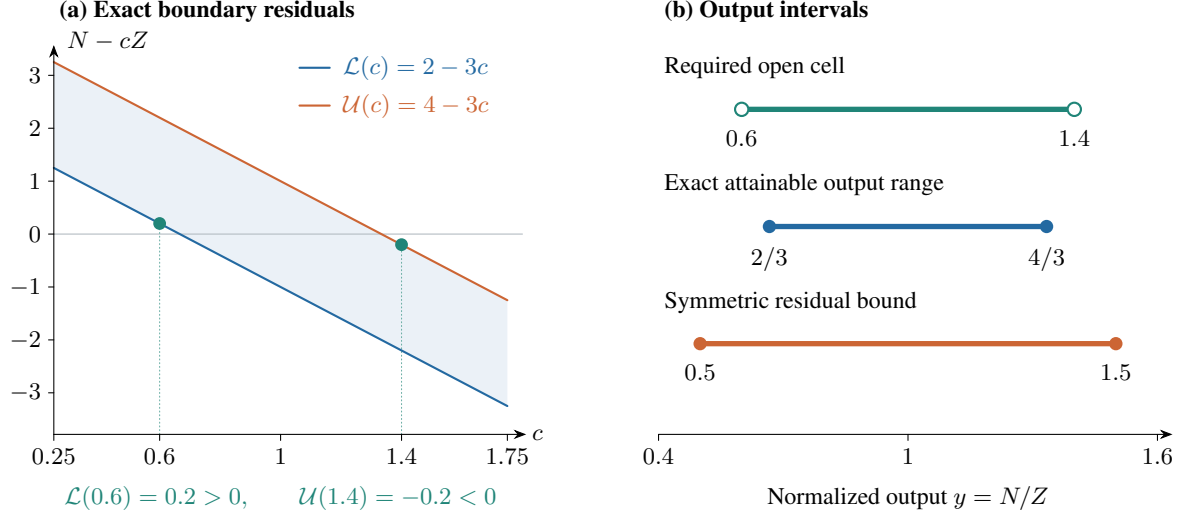


Figure 5. Boundary residuals for Example A.1. The attained output range is closed; the certified open cell is illustrative, not a BF16 rounding cell.

A.7. Joint Value Coupling

Example A.2. For $z \in [1, 4]$, $m \in [1/2, 3/2]$, and $-z/2 \leq m \leq z/2$, consider $m - z/4$. The box gives $[-1/2, 5/4]$; the cone with the mass interval gives $[-3, 1]$. Intersecting these scalar bounds gives $[-1/2, 1]$, whereas joint optimization gives $[-1/2, 3/4]$. This compares metadata geometry without asserting token realizability of every vertex.

A.8. Margin and Refinement

The residual width separates mass and numerator uncertainty:

$$\begin{aligned} W_j(c) &= \mathcal{U}_j(c) - \mathcal{L}_j(c) \\ &= \sum_b [m_{bj}^+ - m_{bj}^- + |\nu_{bj} - c|(u_b - l_b)]. \end{aligned} \quad (25)$$

Proposition A.3 (Margin versus uncertainty). *For $a_j < y_j < b_j$, certification follows if*

$$\begin{aligned} W_j(a_j) &< Z(y_j - a_j), \\ W_j(b_j) &< Z(b_j - y_j). \end{aligned}$$

Proof. The lower endpoint is at least the true residual minus its enclosure width; the upper endpoint is at most the true residual plus its width. The proposed inequalities therefore give both strict signs. $\square$

This explanatory condition requires unknown truth, so is not an executable test. It explains difficult near-midpoint coordinates and prioritizing large width contributions. If sound widths converge to zero and the output stays strictly inside a fixed cell, that cell eventually passes. The bounded executor does not assume convergence before fallback.

A.9. Diagonal Retention Need Not Produce Nested Intervals

Retaining the diagonal weakly reduces the row-sum witness but shifts the approximation center, so the resulting intervals need not be nested. For $H_{11} = 9$, $H_{23} = H_{32} = 10$, all other entries zero, and $u = (1, 0, 0)$, both witnesses equal 10. Retaining the diagonal changes $[-10, 10]$ to $9 + [-10, 10]$. Acceptance need not improve for every cell; both representations are exposed by the implementation.

A.10. Remainder and Projection Growth

Figure 6 compares the sharp remainder coefficient with the Lagrange bound and isolates the separate cost of discarded-score variation.

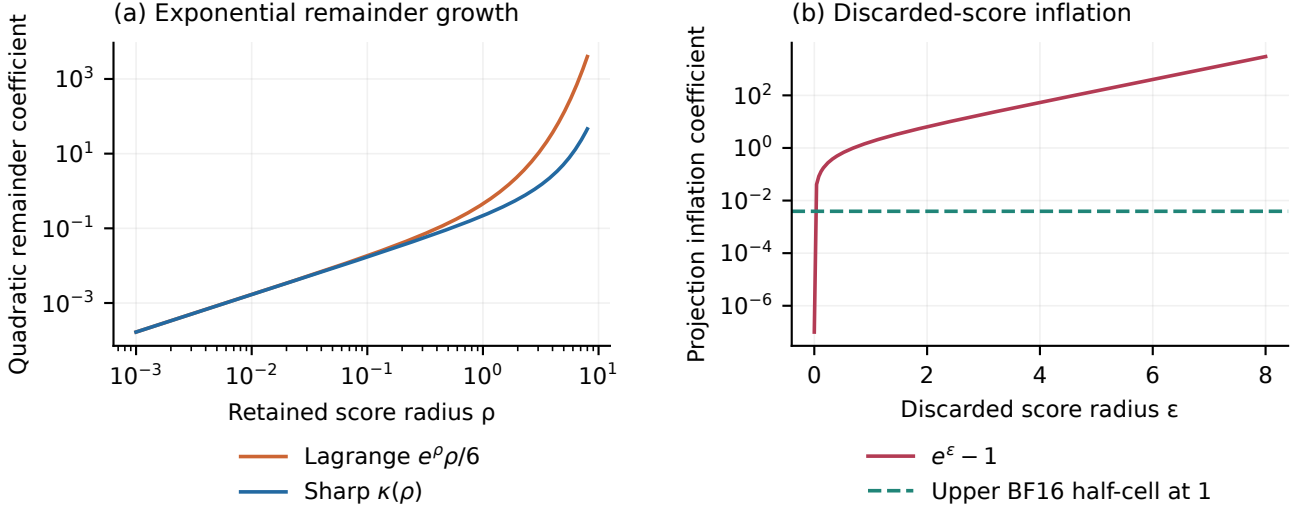


Figure 6. The sharp coefficient improves the Lagrange constant but still grows exponentially. Discarded scores impose a separate exponential cost. The BF16 line is a unit-scale guide, not a sufficient certificate.

B. Formalization and General Consumers

Lean 4.24.0 and the pinned mathlib revision build all 64 local theorems. The complete axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`. The [audit manifest](#) records source hashes and dependencies for every declaration; the [formalization map](#) gives the complete list.

The composed theorem. The observation theorem in Table 1 applies derived block bounds to the actual finite, per-token real exponential quotient. Its witness assumes nonempty tokens, centering, nonnegative radii, radius bounds, retained-tensor symmetry, and an absolute row-sum bound on the omitted signed tensor. It does not assume the Taylor remainder or the final enclosure. Lean uses finite sums where the manuscript uses means: for a block of size n , its H , D , and η are multiplied by n .

Table 1. Key declarations in the built Lean proof chain. The complete audit covers 64 theorems.

Declaration	Established property
<code>exp_remainder_sharp</code>	Actual exponential-series remainder bound.
<code>quadratic_rowsum_bound</code>	Symmetric omitted-tensor bound from absolute row sums.
<code>finite_mean_centering</code>	Exact means supply finite-sum centering.
<code>second_moment_contraction</code>	Projected second-moment contraction.
<code>signed_moment_contraction</code>	Signed key/value tensor contraction.
<code>full_score_enclosure</code>	Full-score block mass and centered-numerator bounds.
<code>full_attention_observation</code>	Strict cell membership for the real exponential quotient.
<code>observation_cell_iff</code>	Necessary-and-sufficient strict signs over the independent box.
<code>centered_value_mass_coupling</code>	Positive-weight value-range/mass dependence.
<code>gated_residual_identity</code>	Exact smooth-gate residual identity.
<code>gated_remainder_bound</code>	Smooth-gate error bound under a Taylor witness.

Executable bridges. The rational exponential, interval conversion, BF16 cell construction, polygon support optimizer, and runtime identity validation are not extracted from Lean. Neither are CUDA memory accesses, instruction rounding, overflow, compiler transformations, or state updates. The concrete IEEE model, including signed zero, infinities, NaNs, and midpoint ties, is not formalized. `monotone_rounding` assumes an abstract monotone real function. The numerical summary path cannot discharge the directed checker’s sound-input premise. These are implementation obligations beyond the real theorem.

B.1. Other Observation Contracts

If a consumer is constant on a set containing the output, its value is fixed (`constant_consumer`). For monotone rounding Q , $l \leq y \leq u$ and $Q(l) = Q(u)$ suffice. For `argmax`, $l_w > u_j$ for all $j \neq w$ certifies winner w (`strict_argmax`). Equality of complete deterministic state transitions implies equality of repeated execution (`transition_composition`); this theorem does not prove an executor’s transition-equality premise.

B.2. Smooth Gates

The residual certificate applies to any positive normalized weights with sound block envelopes. Signed unnormalized contractions instead sum intervals. For a smooth gate, let $p = \phi(t + \alpha)$, $p_0 = \phi(t)$, and $p_1 = \phi'(t)$. Then

$$\begin{aligned} p(s + \gamma) - (p_0 s + p_1 \alpha s + p_0 \gamma) \\ = p_1 \alpha \gamma + (p - p_0 - p_1 \alpha)(s + \gamma). \end{aligned} \quad (26)$$

If $|\alpha| \leq \rho_g$, $|\gamma| \leq \rho_u$, and $|p - p_0 - p_1 \alpha| \leq M \rho_g^2 / 2$, then

$$|\text{error}| \leq |p_1| \rho_g \rho_u + \frac{1}{2} M \rho_g^2 (|s| + \rho_u). \quad (27)$$

Lean proves the identity and bound; Table 1 lists their declarations. A curvature witness for a concrete gate remains a separate obligation. For SwiGLU, absolute downstream weights multiply these nonnegative radii before summation. Contracted coefficients may save an intermediate while requiring more storage than the original weights; no feed-forward speedup is established here.

C. Validation and Experimental Protocols

C.1. Arithmetic and Implementation Validation

The successful Lean 4.24.0 build and axiom audit cover 64 local declarations, including the exponential remainder, finite signed tensor contractions, full-score block enclosure, and composed real-attention quotient. The only axioms are `propext`, `Classical.choice`, and `Quot.sound`; [Lean audit manifest](#) records source hashes and per-theorem dependencies. The polygon optimizer, rational exponential routine, BF16 cell constructor, state identity, and CUDA instruction semantics remain separate implementation obligations.

The 46-test CPU suite includes randomized enclosure tests, 500 exact polygon comparisons with an independent half-plane intersection implementation, 40 direct 100-digit attention comparisons across rank and refinement, and 40 numerical/exact geometry comparisons. Invalid dimensions, nonintegral indices, negative bounds, unsupported domains, stale source/query/shift state, and midpoint behavior are covered.

This audit found an arithmetic defect: integer endpoints allowed reciprocal evaluation through Python’s floating-point `1 / int`, so the upper endpoint of $[1, 1]/[3, 3]$ could lie below $1/3$. Endpoints and scale factors now normalize to exact fractions; the regression requires both endpoints to equal $1/3$ exactly.

The imported CUDA checker retains all 74 rational-certified rows among 160 rationally generated rows, certifies no rationally refused row in that fixture, and rejects 13 invalid-domain or strict-boundary controls. Numerical fixtures compare the original, shared-scalar, and parallel implementations and dense block correction against CPU results. Memory, race, and synchronization checks report no errors on the exercised numerical paths; imported CUDA memory checking also reports no errors. These finite checks complement the theorem chain without establishing instruction-level correctness for all inputs.

C.2. Exact-Rational Coupling Checks

Independent high-precision coverage comparisons use 110 decimal digits and a scaled 10^{-95} tolerance to compare against exact zero-width intervals. An initial zero-tolerance attempt failed through cancellation in a constant channel and is documented. Certificate and rational-oracle comparisons use exact fractions without this tolerance. Stored decimal endpoints are display approximations; broad-channel failures and exact-midpoint abstentions remain in the artifact.

Table 2. Initial exact-rational coupling results. Denominators count coordinates, not independent model examples.

Result	Count
Box certificate accepts	57/149
Coupled certificate accepts	61/149
Global hull accepts	50/149
Narrower coupled residual	30/149
Unchanged residual width	119/149
Accepts beyond global hull	1
False certificate / coverage failure	0

C.3. Uncertainty Localization and GPU Controls

The CPU experiment in Section 5.1 isolates correction work. Figure 7 shows the selected token fractions for every query before the GPU controls below compare complete-output coverage and cost.

GPU synthetic controls use $d = h = 64$ and BF16 input values represented exactly in binary64 on the summary path. Tight clusters can pass completely. Moderate cases pass 10,197 of 19,008 coordinates across configurations but none of 297 complete outputs; these aggregates reuse seeded inputs across ranks. The centered-value control passes 1,986 of 2,048 coordinates but only 6 of 32 outputs. Complete-output acceptance is a conjunction of coordinate decisions, so high marginal coverage need not yield useful output coverage. This observation assumes no coordinate independence. Exact-midpoint controls reject every strict cell by design. In the measured GPU path, one failed coordinate triggers full-batch fallback (Figure 8).

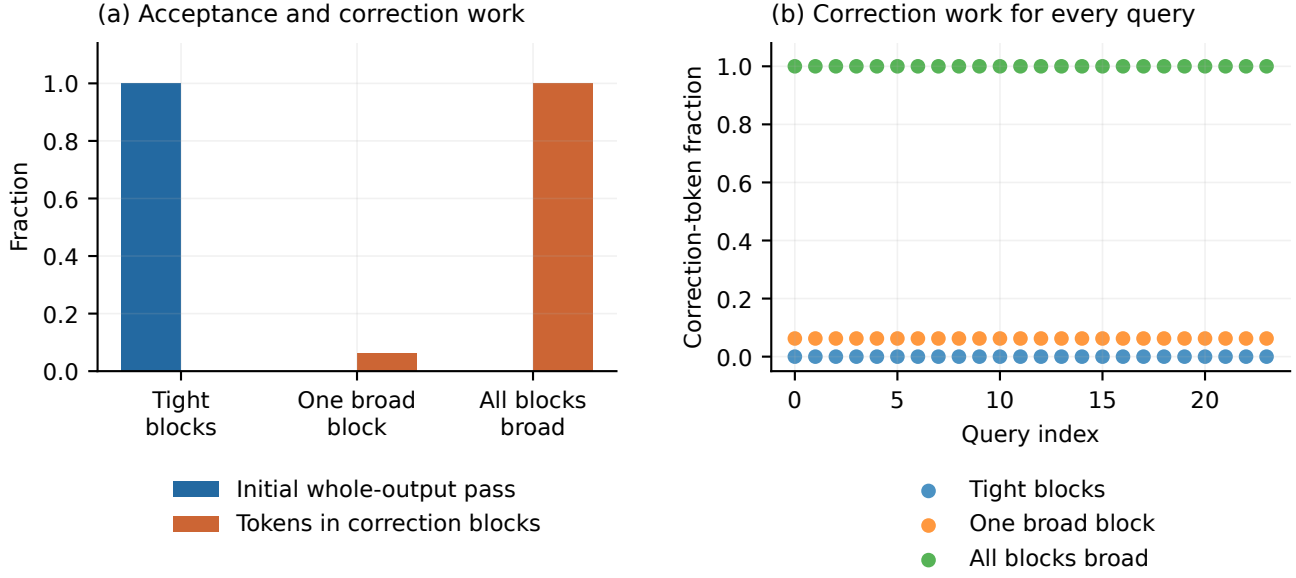


Figure 7. Correction-token fractions for every query in tight, mixed, and broad CPU inputs. Local uncertainty permits local correction; diffuse uncertainty requires all tokens. Full-source validation reads are additional and included in CPU timings.

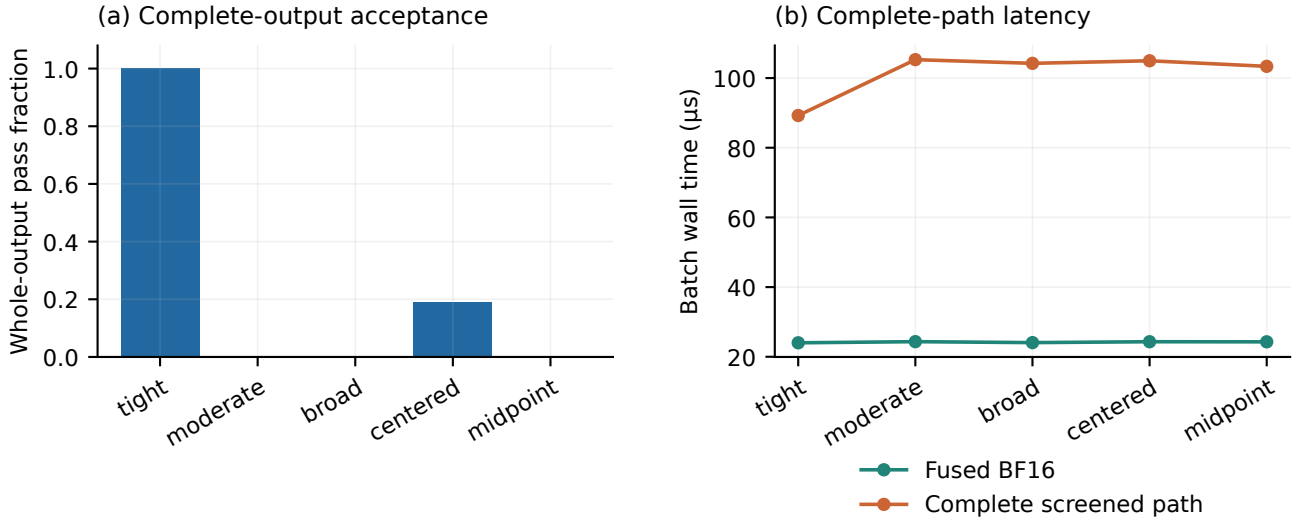


Figure 8. Whole-output coverage and complete GPU costs for rank-four, $N = 8192$, $Q = 32$ controls varying uncertainty and cell margins. The cost includes the full-batch fallback triggered by any failed coordinate.

C.4. Trained-Activation Capture and Diagnostics

No coordinate or whole output passes the numerical screen on either trained activation set. Rank eight covers all 24 layers and 14 query heads over six prefixes: 2,016 head instances and 129,024 coordinates. Full rank covers layers 0, 12, and 23, adding 252 head instances. We capture actual post-RoPE Q/K/V arguments at the attention call of Qwen2.5-0.5B (Qwen Team, 2024), pinned revision 060db64 (the full commit hash is recorded in the [capture manifest](#)), using Transformers 4.51.3 and BF16 weights. Three deterministic original texts cover prose, code, and arithmetic at prefix lengths 1,024 and 4,096. The final prefill query sees the entire prefix; contiguous GQA groups contain seven query heads per KV head, and scaling by 1/8 is exact. This is an activation diagnostic, not a language-model accuracy benchmark or production-traffic sample.

Table 3. Medians of trained-activation radius and cell-margin diagnostics. Full rank uses the three representative layers.

Diagnostic	Rank 8	Rank 64
Maximum-block retained bound ρ	1.32	29.91
Maximum-block discarded bound ϵ	34.24	0
Actual maximum projected score deviation	0.85	8.91
Minimum-channel BF16 boundary margin	8.95×10^{-7}	6.66×10^{-7}

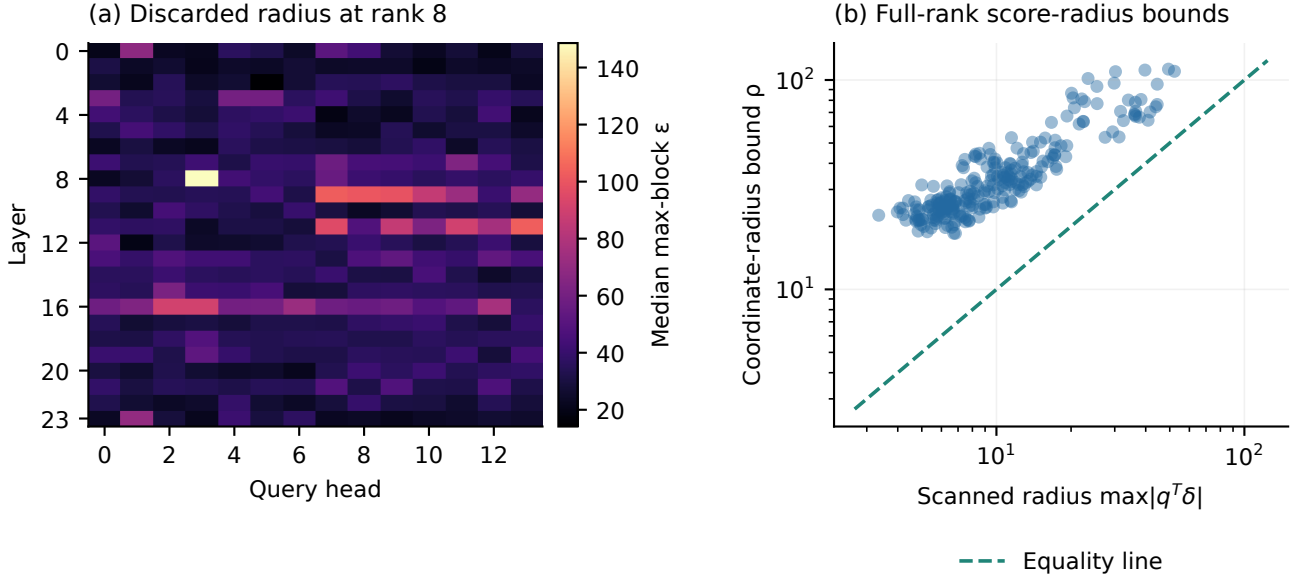


Figure 9. Trained-model score radii by layer and head. Low rank leaves large discarded-score variation; full-rank coordinate bounds exceed scan-derived radii, while actual score variation remains substantial. Computing scanned diagnostic radii reads tokens and is not free query metadata.

C.5. GH200 Environment and Timing Protocol

The final sweep contains 57 synthetic and nine actual prose trace/rank configurations on one NVIDIA GH200: compute capability 9.0, 97,871 MiB reported GPU memory, driver 580.105.08, CUDA 12.8, and PyTorch 2.7.0. Synthetic cases vary $N \in \{1024, 8192, 32768\}$, rank, spread, and $Q \in \{1, 32\}$. Here $Q = 32$ is a noncausal query batch sharing K/V, not independent serving requests. Trace cases use seven query heads sharing one KV head at the final prefix.

The baseline explicitly forces PyTorch’s fused FlashAttention SDPA backend (Dao et al., 2022), matching BF16 input values, scale, visible keys, and zero dropout. A separate binary64 softmax/matmul computation checks outputs. Seven samples of 20 invocations provide raw CUDA-event and host-wall times; plotted quantiles summarize those samples, not workload confidence intervals. Resident timings exclude setup; construction, uploads, casts, allocations, workspace, and graph capture are recorded separately. Complete eager timings include screening, device decision reduction, CPU synchronization, and either an output cast or executed full-batch fallback (Figure 10).

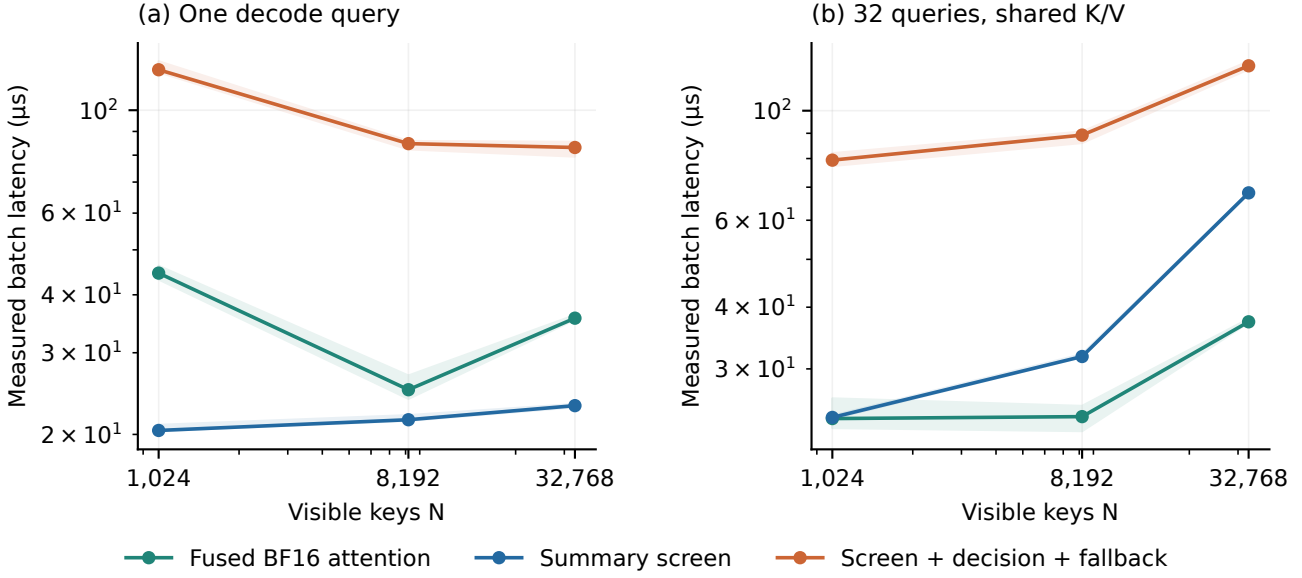


Figure 10. GH200 resident screening and complete-path latency against fused dense attention for tight rank-four inputs. Wall-clock medians and 10th–90th sample quantiles include each complete path’s actual decisions and fallback behavior.

The eager timing helper performs three warmup calls and synchronizes before sampling. Each sample brackets 20 invocations with CUDA events, waits for the ending event, and divides both event elapsed time and host-wall elapsed time by 20. Host samples therefore include Python launch and synchronization costs. Graph measurements first warm the operation on a separate stream, capture 20 invocations with fixed inputs, then warm graph replay three times and collect seven replay samples. Reported replay times are divided by 20; graph capture time is stored separately. No adaptive scheduler or fallback is captured in the replay graph.

All 66 recorded cases use seed 902105. The 57 synthetic configurations contain 19 tight, 18 moderate, 18 broad, one centered-value, and one exact-midpoint case. Seeded input generation is repeated across retained ranks, so aggregate acceptance counts reuse inputs. The nine GPU trace cases are three prose captures, from layers 0, 12, and 23 at prefix length 4,096, evaluated at ranks 0, 4, and 16. They are separate from the rank-eight and full-rank CPU activation diagnostics.

C.6. Kernel Ablation, Setup, and Footprint

Even the favorable $N = 32768, Q = 1, r = 4$ case loses end to end. Its resident parallel screen takes 20.09 μs under fixed-input graph replay versus dense attention’s 31.30 μs ; the complete eager path takes 83.11 μs versus 35.62 μs , plus about 51.50 ms of reusable summary/workspace setup. CPU moment-and-extrema arrays occupy 1,000,448 bytes, the uploaded box summary 869,376 bytes, and original BF16 K/V 8,388,608 bytes. These payloads exclude 262,144 bytes of group indices and Python object/hash overhead; scratch and retained original K/V are reported separately. Neither serving-system memory savings nor a complete-path speedup follows. Every measured complete path is slower than dense, so no finite reuse count repays setup under these costs (Figure 12).

Table 4. Matched graph replay medians in microseconds for tight rank-four inputs. All three screening implementations and the dense baseline are retained, including the adverse small-input case.

N	Q	Original	Shared scalar	Parallel	Dense
1024	32	20.47	19.87	20.84	9.80
8192	1	35.69	35.60	18.72	14.55
32768	1	118.65	118.14	20.09	31.30
32768	32	143.28	147.52	64.89	34.21

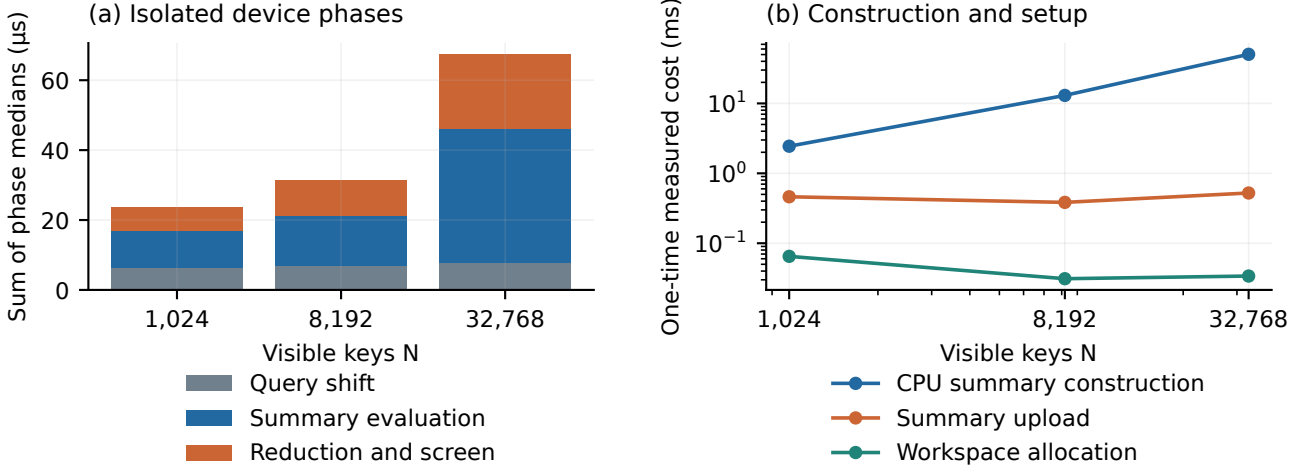


Figure 11. Isolated GPU phase medians and summary setup costs. CPU construction, upload, and workspace allocation remain explicit; complete latency is measured separately rather than reconstructed from phase sums.

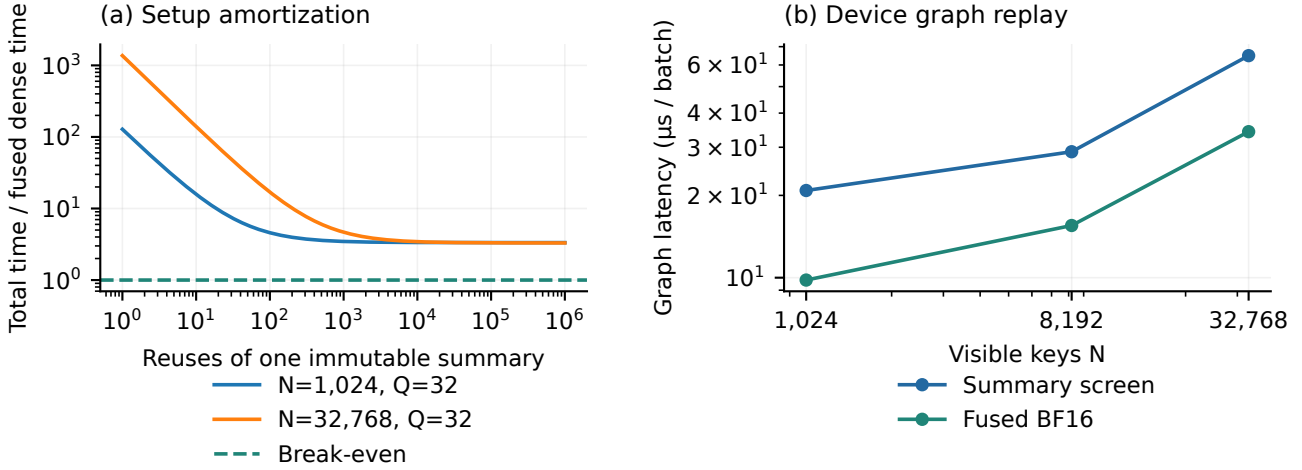


Figure 12. Setup amortization and matched graph device costs. No measured complete path has a finite setup break-even. Fixed-input graph replay omits the adaptive host decision and constitutes a separate experiment.

C.7. Retained Failures and Same-Machine Reproduction

The first trained-trace parity attempt failed nine absolute-residual tolerance checks because broad bounds had extremely large magnitudes. A scaled 10^{-12} comparison addresses this diagnostic scaling while still requiring identical screen decisions and independent output checks; the maximum observed scaled variant difference is 1.33×10^{-15} . The nine failures and original scalar-only sweep are preserved. This tolerance changes implementation-parity diagnostics, not certificate bounds. Figures and tables retain the first complete sweep. A second GH200 run measures commit d2c3668, including final host ABI guards and formatted sources. [GH200 provenance manifest](#) records source, library, and trace hashes; both original and reproduction measurements remain available. All 66 cases pass validation again with unchanged coverage and mismatch counts. Parallel screen graph medians change by -1.08% to $+1.42\%$ across cases, with median ratio 1.002. Complete-path and dense host-wall medians are noisier, with median ratios 1.075 and 1.112. Every complete screened path remains slower than dense, with no finite setup break-even. This reproduces the conclusions on the same machine, without estimating variation across devices, model families, or serving systems.

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